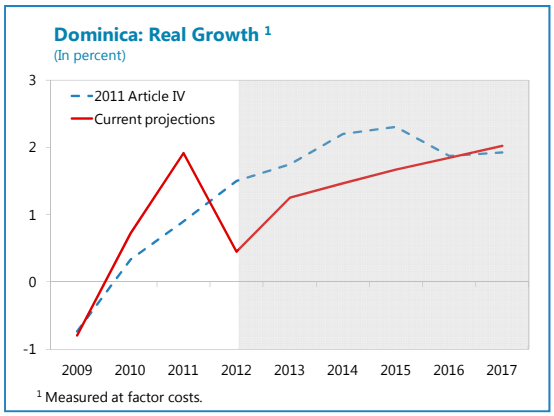
ANNEX II. DEBT SUSTAINABILITY ANALYSIS: UPDATE

The fiscal outlook has deteriorated since the previous reports on weaker outturns and robust investment projections, setting public debt on an upward path under unchanged policies. While the risk of public external debt distress was maintained at "moderate", the odds of a "high" risk of distress are increasing as a result of the deterioration in debt dynamics and the ensuing breach of some debt and debt service thresholds in the baseline and stress scenarios.

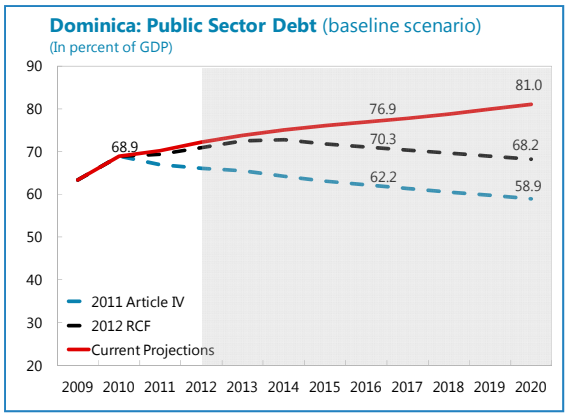
Background. The macroeconomic and fiscal outlook for Dominica has weakened relative to the 2011 Article IV consultations and the 2012 Rapid Credit Facility (RCF) arrangement. This reflects signs of slowing economic activity in 2012 that is expected to affect the near-term outlook; weaker fiscal outcomes in fiscal years 2010–11; and a weaker fiscal outlook in staff's medium-term projections due to a stronger public investment assumption. The assumptions used in the DSA are as follows:

- *Macroeconomic assumptions***.** Growth is projected to slow to about ½ percent in 2012, and to pick up only gradually over the medium term to a potential of close to 2 percent. The potential growth is similar to the one estimated in previous reports, but the near-term prospects are weaker in light of the faltering demand at home and abroad. Inflation is projected to remain around 2 percent over the near and medium-term, similar to the previous reports.



- *Fiscal assumptions.* In staff's baseline scenario, the primary balance improves somewhat over the medium term, due to the compression in infrastructure spending with the winding down of some of the post-flood rehabilitation work. A stronger improvement, however, is held back by (i) a likely fading of non-tax receipts from the *economic citizenship program,* which surged in recent years following unrest in the Middle East; and (ii) the new assumption that large ongoing infrastructure projects would be replaced by new spending once they wind up rather than reduce capital spending *pro tanto*, as in previous reports. Concessional lending and grants are expected to continue playing a major role in financing the overall fiscal deficit in line with Dominica's long-standing policy, with the grant element of new external financing expected at about 27 percent in the coming years, broadly in line with the previous assumptions. The discount rate used to derive the net present value of debt stocks is set to 3 percent, compared to 4 percent previously.

Public debt outlook. The debt outlook has deteriorated since the 2011 Article IV consultation and the RCF request, reflecting both the weaker fiscal position and the downward revision of the near-term growth. Public debt is now projected to remain on an upward path under unchanged policies, compared to a slight decline in previous reports. It increases from 70.2 percent of GDP in FY2011 to above 80 percent by FY2020, away from the ECCU target of 60 percent (Figure A1 and Tables A1-A2). A return to the authorities' primary surplus target of 2.4 percent of GDP—which is also in line with historical primary surpluses achieved over the past decade—would significantly improve the debt outlook, setting it on a rapidly declining path and allowing the authorities to meet the ECCU target even in the presence of plausible macroeconomic shocks.



External debt outlook. The outlook for the external debt, on the other hand, has improved notably over the near and medium-term because of large revisions to the external current account balance, which show a narrower deficit for 2010–11.[1] As a result of these revisions, near-term external debt is projected to be significantly lower than at the time of the RCF, although growing public debt will put external indebtedness on an upward path over the long-term (Table A3). Bound tests show that under the current baseline scenario one of the five thresholds applicable to public external debt is breached (present value of debt-to-GDP), with the other ratios below the thresholds for the entire projection horizon (Figure A2). Under alternative scenarios, the country appears vulnerable to the possible deterioration in the terms of new borrowing, which leads to a breach of another threshold (debt service to exports). In practice, however, the vulnerability is muted by the fact that most of the financing is not on market terms and hence unlikely to be subject to term revisions.

Debt distress rating. Staff has maintained its assessment of a "moderate" risk of public external debt distress, but the odds of a "high" risk of distress are increasing as a result of the deterioration of the debt outlook, the breach of a threshold under the no-policy-change scenario (external public debt-to-GDP ratio), although this is expected only in the long-term, and the breach of additional thresholds under the stress scenarios (external public debt-to-GDP ratio and debt service to export ratio).

[1] A recent tourism expenditure survey by the authorities conducted in August 2010 and used to estimate tourism receipts for the period 2010–11, found a 26 percent increase in tourism spending relative to the previous survey. This, together with a revision in other service receipts, resulted in a 3½ percentage points of GDP reduction in the current account deficit for 2010. This revision, and a less-than-anticipated impact of food and fuel prices on the balance of payments, reduced the current account estimate for 2011 from 23½ percent of GDP at the time of the RCF request to 12¾ percent of GDP currently.

Table A1. Dominica: Public Sector Debt Sustainability Framework, Baseline Scenario, 2009–2032

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2009	2010	2011	Average 5/	Standard Deviation 5/	2012	2013	2014	2015	2016	2017	2012-17 Average	2022	2032	2018-32 Average
Public sector debt 1/	63.3	68.9	70.2			72.3	73.7	75.0	76.0	76.9	77.7		83.5	95.7	
o/w foreign-currency denominated	44.3	50.5	50.2			52.4	53.7	54.1	54.8	55.3	55.6		57.5	66.1	
Change in public sector debt	-1.0	5.6	1.3			2.1	1.5	1.2	1.0	0.8	0.8		1.2	1.3	
Identified debt-creating flows	-0.9	6.1	3.9			2.1	1.5	1.2	1.0	0.8	0.8		1.2	1.3	
Primary deficit	-1.5	4.3	3.0	-2.3	3.5	2.5	2.0	1.5	1.6	1.6	1.4	1.8	1.4	1.7	1.5
Revenue and grants	42.9	43.9	37.8			36.4	36.1	35.9	35.7	35.5	35.7		35.5	35.2	
of which: grants	8.7	9.7	4.8			4.0	4.0	4.0	4.0	4.0	4.0		4.0	4.0	
Primary (noninterest) expenditure	41.4	48.2	40.8			38.9	38.1	37.4	37.3	37.2	37.0		36.9	36.9	
Automatic debt dynamics	0.6	1.8	0.8			-0.4	-0.5	-0.3	-0.6	-0.8	-0.5		-0.2	-0.5	
Contribution from interest rate/growth differential	1.8	0.9	0.8			0.3	-0.3	-0.4	-0.6	-0.8	-0.8		-0.7	-0.9	
of which: contribution from average real interest rate	1.0	1.6	1.5			0.6	0.5	0.7	0.6	0.6	0.7		0.8	0.8	
of which: contribution from real GDP growth	0.8	-0.8	-0.7			-0.3	-0.9	-1.1	-1.2	-1.4	-1.5		-1.5	-1.7	
Contribution from real exchange rate depreciation	-1.2	1.0	0.0			-0.7	-0.2	0.1	0.1	0.0	0.3		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-0.1	-0.5	-2.6			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other Sustainability Indicators															
PV of public sector debt	...	...	60.8			62.4	63.5	65.2	65.7	66.0	67.3		73.7	85.0	
o/w foreign-currency denominated	...	...	40.7			42.6	43.4	44.3	44.4	44.5	45.3		47.7	54.8	
o/w external	...	...	40.7			42.6	43.4	44.3	44.4	44.5	45.3		47.7	54.8	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	3.5	8.9	9.2			9.0	7.9	8.5	7.9	7.9	7.6		10.8	12.3	
PV of public sector debt-to-revenue and grants ratio (in percent)	…	…	160.7			171.7	175.7	181.5	183.9	185.8	188.7		207.5	241.7	
PV of public sector debt-to-revenue ratio (in percent)	…	…	184.3			192.8	197.6	204.2	207.1	209.4	212.5		233.8	269.4	
o/w external 3/	…	…	123.5			131.6	135.3	138.9	140.0	141.0	142.9		151.4	175.9	
Debt service-to-revenue and grants ratio (in percent) 4/	11.7	10.6	16.3			17.9	16.3	19.5	17.7	17.7	17.5		26.5	30.2	
Debt service-to-revenue ratio (in percent) 4/	14.7	13.6	18.7			20.1	18.3	21.9	19.9	20.0	19.7		29.8	34.0	
Primary deficit that stabilizes the debt-to-GDP ratio	-0.5	-1.3	1.8			0.4	0.5	0.3	0.6	0.8	0.5		0.2	0.5	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	-1.3	1.2	1.0	2.8	3.6	0.4	1.3	1.5	1.7	1.8	2.0	1.4	1.8	1.9	1.9
Average nominal interest rate on forex debt (in percent)	3.1	2.0	1.8	3.1	1.0	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8
Average real interest rate on domestic debt (in percent)	0.3	4.8	5.0	3.6	2.5	2.3	3.2	3.8	3.6	3.4	3.9	3.4	3.9	3.9	3.8
Real exchange rate depreciation (in percent, + indicates depreciation)	-2.6	2.2	0.0	0.7	2.1	-1.5	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	3.7	-1.8	0.6	1.3	2.1	3.0	2.3	1.8	1.9	2.0	1.5	2.1	1.1	1.3	1.2
Growth of real primary spending (deflated by GDP deflator, in percent)	0.0	0.2	-0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Grant element of new external borrowing (in percent)	...	...	...	…	…	26.5	27.0	26.3	25.4	24.7	24.4	25.7	24.8	24.4	...

Sources: Country authorities; and staff estimates and projections.

1/ Gross debt of the central government, including nonfinancial public sector debt guaranteed by the central government, but excluding debt incurred under the Petrocaribe arrangement with Venezuela. Revenues and expenditures include imputations for the rest of the public sector.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table A2. Dominica: Sensitivity Analysis for Key Indicators of Public Debt 2012–2032

	Projections							
	2012	2013	2014	2015	2016	2017	2022	2032
PV of Debt-to-GDP Ratio								
Baseline	62	63	65	66	66	67	74	85
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	62	58	56	52	48	46	33	9
A2. Primary balance is unchanged from 2012	62	64	67	68	69	71	83	103
A3. Permanently lower GDP growth 1/	62	64	67	69	71	74	93	153
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2013-2014	62	65	70	72	74	76	90	114
B2. Primary balance is at historical average minus one standard deviations in 2013-2014	62	63	64	65	65	66	73	84
B3. Combination of B1-B2 using one half standard deviation shocks	62	61	61	62	63	64	72	86
B4. One-time 30 percent real depreciation in 2013	62	81	83	83	83	84	91	106
B5. 10 percent of GDP increase in other debt-creating flows in 2013	62	73	75	75	76	77	83	94
PV of Debt-to-Revenue Ratio 2/								
Baseline	172	176	181	184	186	189	208	242
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	172	162	156	146	137	129	95	26
A2. Primary balance is unchanged from 2012	172	177	185	190	194	200	235	293
A3. Permanently lower GDP growth 1/	172	178	186	192	198	206	260	426
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2013-2014	172	181	194	201	206	213	251	323
B2. Primary balance is at historical average minus one standard deviations in 2013-2014	172	174	179	181	183	186	205	239
B3. Combination of B1-B2 using one half standard deviation shocks	172	169	170	174	176	180	202	243
B4. One-time 30 percent real depreciation in 2013	172	225	231	232	233	235	255	301
B5. 10 percent of GDP increase in other debt-creating flows in 2013	172	202	208	211	212	215	233	266
Debt Service-to-Revenue Ratio 2/								
Baseline	18	16	19	18	18	18	26	30
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	18	16	19	16	16	16	15	6
A2. Primary balance is unchanged from 2012	18	16	20	18	18	18	29	36
A3. Permanently lower GDP growth 1/	18	16	20	18	18	18	30	48
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2013-2014	18	17	20	19	19	19	31	39
B2. Primary balance is at historical average minus one standard deviations in 2013-2014	18	16	19	18	18	17	26	30
B3. Combination of B1-B2 using one half standard deviation shocks	18	16	19	17	17	17	25	30
B4. One-time 30 percent real depreciation in 2013	18	18	25	22	22	22	38	52
B5. 10 percent of GDP increase in other debt-creating flows in 2013	18	16	20	19	18	18	32	33

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

Table A3a. Dominica: External Debt Sustainability Framework, Baseline Scenario, 2009–2032[1/]

(In percent of GDP, unless otherwise indicated, fiscal year basis)

	Actual			Historical Average [2/]	Standard Deviation [2/]	Projections									
	2009	2010	2011			2012	2013	2014	2015	2016	2017	2012-2017 Average	2022	2032	2018-2032 Average
External debt (nominal) 1/	**68.2**	**72.8**	**71.1**			**73.9**	**72.8**	**71.6**	**71.1**	**70.7**	**70.0**		**71.3**	**78.5**	
o/w public and publicly guaranteed (PPG)	44.3	50.5	50.1			52.4	53.7	54.1	54.8	55.3	55.6		57.5	66.1	
Change in external debt	1.3	4.6	-1.6			2.7	-1.1	-1.2	-0.5	-0.4	-0.6		0.5	0.7	
Identified net debt-creating flows	11.0	11.3	7.7			9.1	7.9	7.4	7.0	7.0	6.9		7.1	6.1	
Non-interest current account deficit	**19.2**	**14.6**	**11.0**	**15.3**	**4.5**	**11.7**	**12.1**	**12.2**	**12.5**	**13.2**	**13.9**		**13.9**	**13.0**	13.9
Deficit in balance of goods and services	22.3	18.1	13.0			14.1	14.5	14.5	14.9	15.5	16.3		16.4	17.1	
Exports	32.3	35.8	39.8			38.3	37.7	37.2	36.8	36.4	36.3		36.8	37.5	
Imports	54.5	53.9	52.8			52.3	52.1	51.7	51.6	51.9	52.6		53.3	54.6	
Net current transfers (negative = inflow)	-4.0	-4.1	-3.4	-4.3	0.6	-3.7	-3.7	-3.7	-3.7	-3.7	-3.8		-4.0	-4.6	-4.2
o/w official	-0.1	-0.1	0.4			0.0	0.0	0.0	0.1	0.1	0.1		0.1	0.0	
Other current account flows (negative = net inflow)	0.9	0.6	1.4			1.3	1.3	1.3	1.3	1.3	1.4		1.5	0.5	
Net FDI (negative = inflow)	**-8.5**	**-5.0**	**-3.6**	**-7.2**	**2.5**	**-3.6**	**-4.6**	**-5.1**	**-5.7**	**-6.3**	**-7.0**		**-7.0**	**-7.0**	-7.0
Endogenous debt dynamics 3/	**0.3**	**1.8**	**0.3**			**1.0**	**0.5**	**0.3**	**0.2**	**0.1**	**0.0**		**0.2**	**0.2**	
Contribution from nominal interest rate	1.9	1.4	1.5			1.3	1.3	1.4	1.4	1.4	1.4		1.4	1.6	
Contribution from real GDP growth	0.8	-0.8	-0.7			-0.3	-0.9	-1.0	-1.2	-1.3	-1.4		-1.3	-1.4	
Contribution from price and exchange rate changes	-2.4	1.2	-0.4			…	…	…	…	…	…		…	…	
Residual (3-4) 4/	**-9.8**	**-6.8**	**-9.3**			**-6.4**	**-9.0**	**-8.6**	**-7.6**	**-7.4**	**-7.5**		**-6.6**	**-5.4**	
o/w exceptional financing	-9.1	-6.9	-3.9			-5.1	-4.7	-4.7	-4.7	-4.7	-4.7		-4.7	-4.7	
PV of external debt 5/	...	...	61.8			64.1	62.5	61.8	60.8	59.8	59.7		61.5	67.8	
In percent of exports	...	...	155.1			167.5	166.0	166.3	165.3	164.2	164.3		167.2	180.7	
PV of PPG external debt	**...**	**...**	**40.7**			**42.6**	**43.4**	**44.3**	**44.4**	**44.5**	**45.3**		**47.7**	**55.4**	
In percent of exports	**...**	**...**	**102.3**			**111.4**	**115.3**	**119.2**	**120.8**	**122.1**	**124.7**		**129.7**	**147.6**	
In percent of government revenues	**...**	**...**	**123.5**			**131.6**	**135.3**	**138.9**	**140.0**	**141.0**	**142.9**		**151.4**	**177.8**	
Debt service-to-exports ratio (in percent)	**13.5**	**9.9**	**10.8**			**11.7**	**10.0**	**13.0**	**11.2**	**24.5**	**24.6**		**32.5**	**20.4**	
PPG debt service-to-exports ratio (in percent)	**11.9**	**8.6**	**9.4**			**10.4**	**8.8**	**11.8**	**9.9**	**9.9**	**9.6**		**17.4**	**19.1**	
PPG debt service-to-revenue ratio (in percent)	**11.3**	**9.0**	**11.4**			**12.3**	**10.3**	**13.8**	**11.5**	**11.4**	**11.0**		**20.3**	**23.0**	
Total gross financing need (Billions of U.S. dollars)	0.1	0.1	0.1			0.1	0.1	0.1	0.1	0.1	0.1		0.1	0.1	
Non-interest current account deficit that stabilizes debt ratio	17.9	10.0	12.6			9.0	13.2	13.4	13.0	13.6	14.5		13.4	12.2	
Key macroeconomic assumptions															
Real GDP growth (in percent)	-1.3	1.2	1.0	2.8	3.6	0.4	1.3	1.5	1.7	1.8	2.0	1.4	1.8	1.9	1.9
GDP deflator in US dollar terms (change in percent)	3.7	-1.8	0.6	1.3	2.1	3.0	2.3	1.8	1.9	2.0	1.5	2.1	1.1	1.3	1.2
Effective interest rate (percent) 6/	2.9	2.0	2.0	2.9	0.8	1.9	1.9	1.9	2.0	2.0	2.0	2.0	2.1	2.1	2.1
Growth of exports of G&S (US dollar terms, in percent)	-3.5	10.4	12.9	5.1	6.6	-0.6	2.0	1.9	2.5	3.0	3.2	2.0	3.3	3.3	3.3
Growth of imports of G&S (US dollar terms, in percent)	-7.9	-1.7	-0.5	5.0	10.2	2.5	3.2	2.4	3.4	4.5	4.9	3.5	3.2	3.5	3.4
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	26.5	27.0	26.3	25.4	24.7	24.4	25.7	24.8	24.4	24.7
Government revenues (excluding grants, in percent of GDP)	34.2	34.2	33.0			32.4	32.1	31.9	31.7	31.5	31.7		31.5	31.2	31.4
Aid flows (in Billions of US dollars) 7/	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
o/w Grants	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
o/w Concessional loans	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			5.5	5.3	5.3	5.1	5.1	5.2		5.7	6.1	5.9
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			56.6	60.1	59.6	60.9	60.4	58.3		52.2	48.0	50.9
Memorandum items:															
Nominal GDP (Billions of US dollars)	0.5	0.5	0.5			0.5	0.5	0.5	0.6	0.6	0.6		0.7	1.0	
Nominal dollar GDP growth	2.4	-0.6	1.6			3.4	3.6	3.3	3.6	3.9	3.6	3.6	3.0	3.2	3.1
PV of PPG external debt (in Billions of US dollars)			0.2			0.2	0.2	0.2	0.2	0.3	0.3		0.3	0.5	
(PVt-PVt-1)/GDPt-1 (in percent)						2.6	1.8	1.8	1.3	1.4	1.9	1.8	1.2	1.9	1.7
Gross workers' remittances (Billions of US dollars)	…	…	…			…	…	…	…	…	…		…	…	
PV of PPG external debt (in percent of GDP + remittances)	...	...	40.7			42.6	43.4	44.3	44.4	44.5	45.3		47.7	55.4	
PV of PPG external debt (in percent of exports + remittances)	...	...	102.3			111.4	115.3	119.2	120.8	122.1	124.7		129.7	147.6	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	9.4			10.4	8.8	11.8	9.9	9.9	9.6		17.4	19.1	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt. Revenues include imputations for the rest of the public sector.
2/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
3/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
4/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
5/ Assumes that PV of private sector debt is equivalent to its face value.
6/ Current-year interest payments divided by previous period debt stock.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table A3b. Dominica: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2012–2032

(In percent)

	Projections							
	2012	2013	2014	2015	2016	2017	**2022**	2032
PV of debt-to GDP ratio								
Baseline	43	43	44	44	44	45	**48**	55
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2012-2032 1/	43	44	45	47	48	50	**56**	70
A2. New public sector loans on less favorable terms in 2012-2032 2/	43	44	46	46	47	48	**55**	75
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	43	44	46	46	46	47	**50**	58
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	43	45	48	48	48	49	**52**	58
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	43	45	47	47	47	48	**51**	59
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	43	43	45	45	45	46	**48**	56
B5. Combination of B1-B4 using one-half standard deviation shocks	43	43	44	44	44	45	**48**	56
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	43	61	63	63	63	64	**68**	78
PV of debt-to-exports ratio								
Baseline	111	115	119	121	122	125	**130**	148
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2012-2032 1/	111	116	122	127	132	137	**152**	186
A2. New public sector loans on less favorable terms in 2012-2032 2/	111	117	123	126	129	134	**151**	201
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	111	115	119	121	122	125	**130**	148
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	111	123	139	141	142	145	**151**	166
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	111	115	119	121	122	125	**130**	148
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	111	115	120	122	123	126	**130**	148
B5. Combination of B1-B4 using one-half standard deviation shocks	111	112	114	115	117	119	**124**	144
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	111	115	119	121	122	125	**130**	148
PV of debt-to-revenue ratio								
Baseline	132	135	139	140	141	143	**151**	178
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2012-2032 1/	132	137	142	147	152	157	**178**	225
A2. New public sector loans on less favorable terms in 2012-2032 2/	132	137	143	146	149	153	**176**	242
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	132	138	145	146	147	149	**158**	186
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	132	140	151	153	154	155	**164**	187
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	132	140	147	148	149	151	**160**	188
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	132	135	140	141	142	144	**152**	179
B5. Combination of B1-B4 using one-half standard deviation shocks	132	134	138	139	140	142	**151**	181
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	132	191	196	198	200	202	**214**	252

Table A3b. Dominica: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2012–2032 (continued)

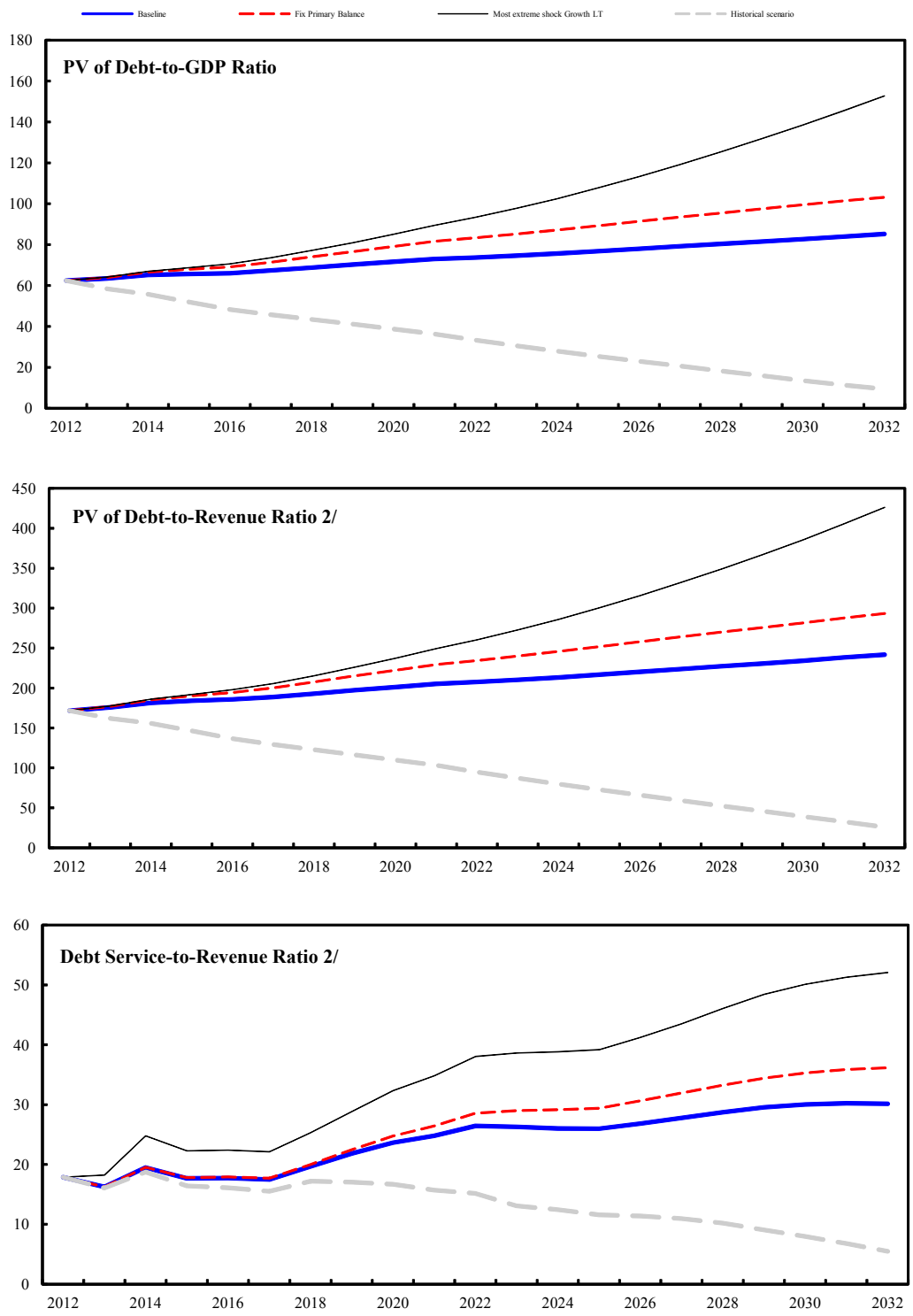
(In percent)

	Projections							
	2012	2013	2014	2015	2016	2017	**2022**	2032
Debt service-to-exports ratio								
Baseline	10	9	12	10	10	10	**17**	19
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2012-2032 1/	10	9	9	10	10	10	**17**	19
A2. New public sector loans on less favorable terms in 2012-2032 2/	10	9	9	10	10	11	**20**	29
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	10	9	9	10	10	9	**17**	19
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	10	9	10	11	11	10	**19**	21
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	10	9	9	10	10	9	**17**	19
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	10	9	9	10	10	9	**17**	19
B5. Combination of B1-B4 using one-half standard deviation shocks	10	9	9	10	10	9	**17**	19
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	10	9	9	10	10	9	**17**	19
Debt service-to-revenue ratio								
Baseline	12	10	14	12	11	11	**20**	23
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2012-2032 1/	12	10	11	11	11	11	**19**	23
A2. New public sector loans on less favorable terms in 2012-2032 2/	12	10	10	11	11	13	**23**	35
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	12	11	11	12	12	11	**21**	24
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	12	10	11	12	12	11	**20**	24
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	12	11	11	12	12	11	**21**	25
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	12	10	11	11	11	11	**20**	23
B5. Combination of B1-B4 using one-half standard deviation shocks	12	11	11	12	11	11	**20**	24
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	12	15	15	16	16	15	**28**	33
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	-7	-7	-7	-7	-7	-7	**-7**	-7

Sources: Country authorities; and staff estimates and projections.

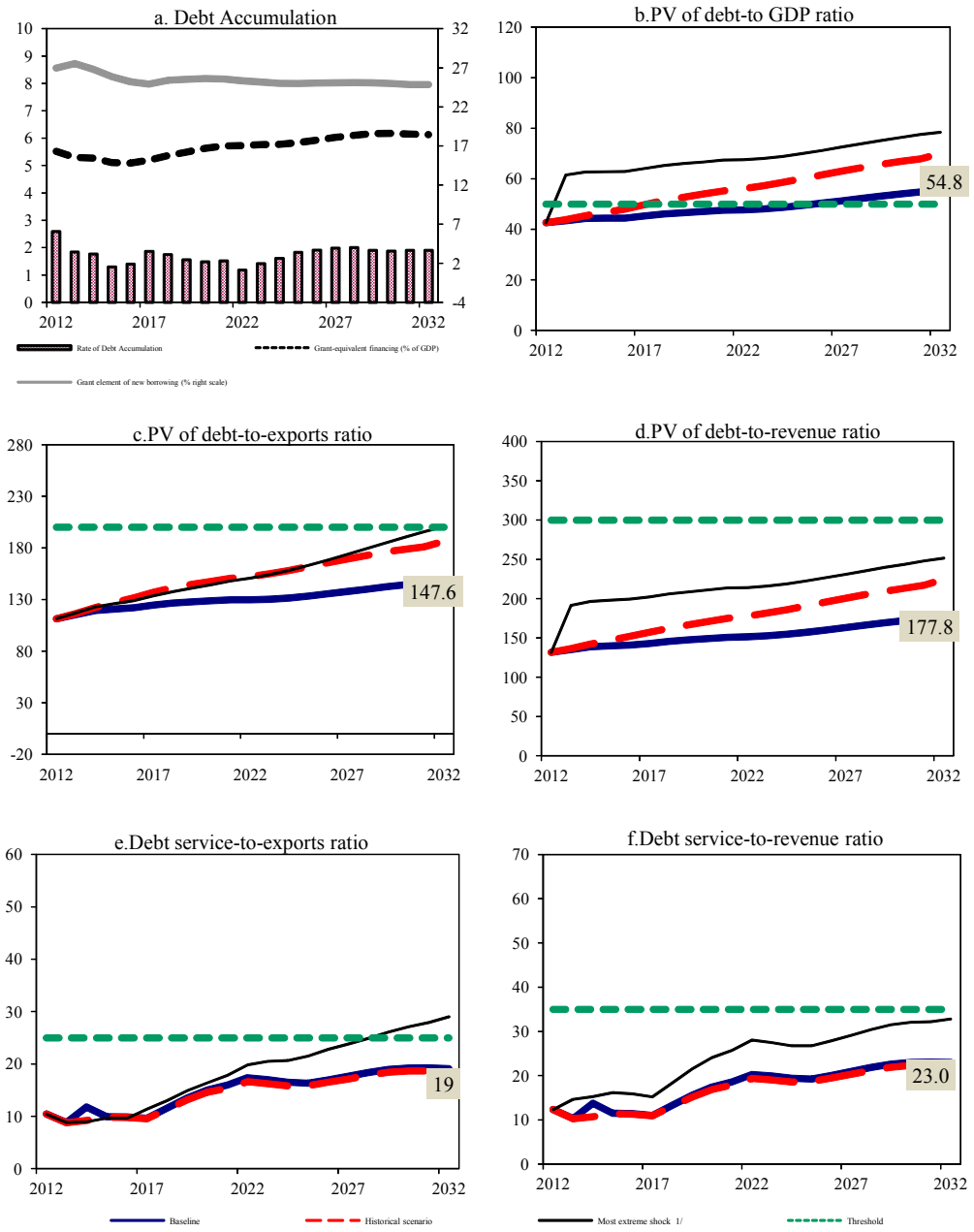
1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Figure A1. Dominica: Indicators of Public Debt Under Alternative Scenarios, 2012–2032[1/]



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2022.
2/ Revenues are defined inclusive of grants.

Figure A2. Dominica: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2012–2032[1/]



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2022. In figure b. it corresponds to a One-time depreciation shock; in c. to a Terms shock; in d. to a One-time depreciation shock; in e. to a Terms shock and in figure f. to a One-time depreciation shock